From Two Loops to a Matrix

A guided Kirchhoff and Jupyter bridge

Student number and last name: _____

Class or period: _____

Learning goals

By the end of this bridge, I can:

- define the three current directions in a two-loop circuit;
- write one KVL equation for each loop and handle the shared resistor;
- rearrange both equations into the same variable order;
- build $A\mathbf{x} = \mathbf{b}$ and solve it numerically and symbolically;
- verify the result with KCL, KVL, and a residual.

Why this is useful

Two loops share one component, so the current through that component depends on both loop currents. A matrix does not replace Kirchhoff's laws. It organizes the equations after the circuit reasoning is complete.

What you will make

You will submit one notebook showing this complete chain:

arrows $\longrightarrow$ two KVL equations $\longrightarrow$ standard form $\longrightarrow$ $A\mathbf{x} = \mathbf{b}$ $\longrightarrow$ two solver methods $\longrightarrow$ checks

Before you start

- Use Question 4 as the worked model, then transfer the method to Question 5.
- Keep the released Question 5 answer closed until your matrix rows read back correctly.
- Download the course notebook from the assignment page.
- Keep resistance in $\mathrm{k}\Omega$, voltage in V, and current in mA.
- This checkpoint is a computational verification activity. No physical lab is required.

Textbook support: Grob, Chapter 9, pp. 264–281. **Practice support:** Problems Manual, Chapter 9.
Engagement review: ElectroBOOM101-008: Kirchhoff Circuit Laws.

Start with current directions and the shared branch

1 Define the variables before writing equations

For the completed Question 4 circuit, use clockwise mesh currents I_1 and I_2. The two mesh currents move through the shared resistor in opposite directions. Define I_3 downward through the shared resistor.

$$I_1 - I_2 - I_3 = 0 \quad \Longrightarrow \quad I_3 = I_1 - I_2$$

This is KCL at the shared branch. It fixes the meaning of every later sign.

I_1	clockwise current assigned to Loop 1
I_2	clockwise current assigned to Loop 2
I_3	downward current through R_3, defined as $I_1 - I_2$

2 Use one sign rule from start to finish

As you walk around a loop:

Source from $-$ to $+$	voltage rise: write $+V_S$
Source from $+$ to $-$	voltage drop: write $-V_S$
Resistor in the current direction	voltage drop: write $-IR$
Resistor against the current direction	voltage rise: write $+IR$

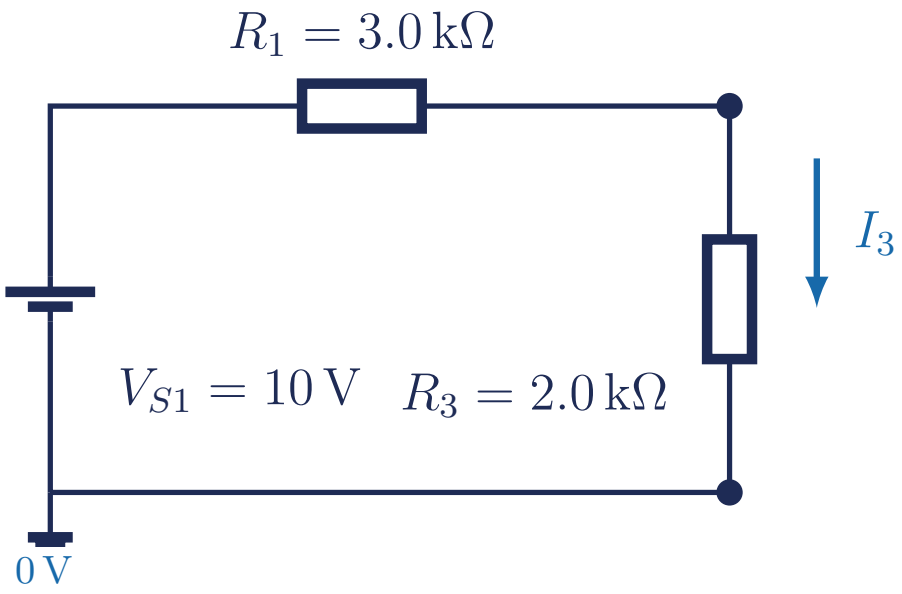
Essential skill: do not jump directly to the matrix. First write the signed KVL terms in the order encountered around each loop. Then substitute $I_3 = I_1 - I_2$.

Quick check before continuing

If the final value of I_3 is negative, what would that mean?

Loop 1: write, substitute, expand, and collect

3 Walk clockwise around Loop 1



Element	Signed KVL term	Reason
$V_{S1} = 10\,\text{V}$	$+10$	rise through the source
$R_1 = 3.0\,\text{k}\Omega$	$-3I_1$	travel with I_1
$R_3 = 2.0\,\text{k}\Omega$	$-2I_3$	travel with the defined I_3

Write the voltage changes before substituting Ohm's law:

$$+10 - V_{R1} - V_{R3} = 0$$

Replace each resistor voltage with IR:

$$V_{R1} = I_1 R_1 = 3I_1, \qquad V_{R3} = I_3 R_3 = 2I_3$$

$$+10 - 3I_1 - 2I_3 = 0$$

Substitute the shared current:

$$+10 - 3I_1 - 2(I_1 - I_2) = 0$$

Distribute -2 to both terms in the bracket:

$$+10 - 3I_1 - 2I_1 + 2I_2 = 0$$

Collect the two I_1 terms:

$$+10 - 5I_1 + 2I_2 = 0$$

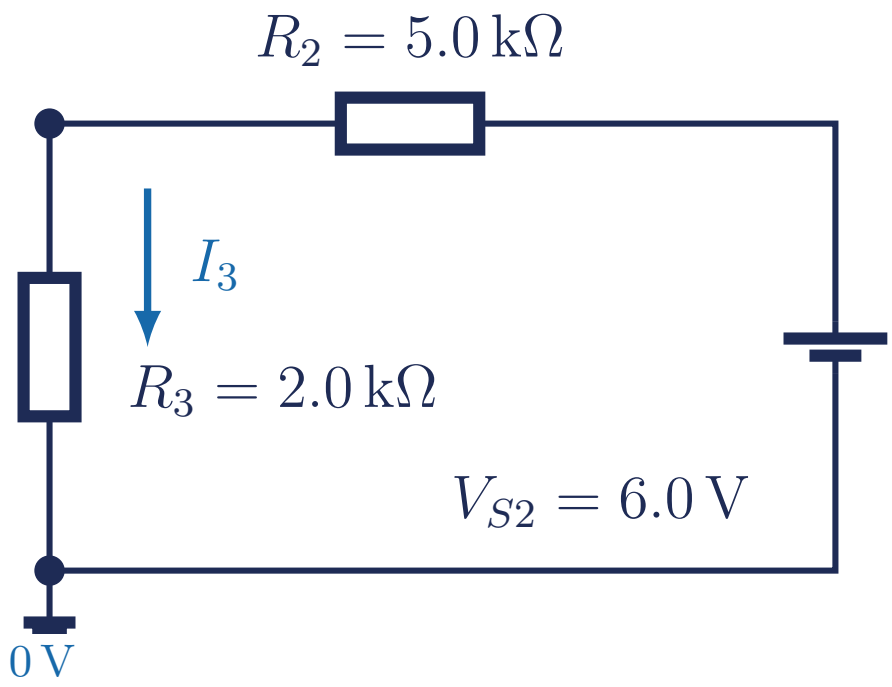
Subtract 10 from both sides, then multiply every term by -1:

$$-5I_1 + 2I_2 = -10 \quad \Longrightarrow \quad \boxed{5I_1 - 2I_2 = 10}$$

The first matrix row is read directly from the standard-form equation: coefficient of I_1, coefficient of I_2, right side. Therefore Row 1 is $[5,\ -2]$ and $b_1 = 10$.

Loop 2: the shared resistor changes sign

4 Walk clockwise around Loop 2



Element	Signed KVL term	Reason
$V_{S2} = 6.0\,\text{V}$	$+6$	rise through the source
$R_2 = 5.0\,\text{k}\Omega$	$-5I_2$	travel with I_2
$R_3 = 2.0\,\text{k}\Omega$	$+2I_3$	travel against the defined I_3

Write the voltage changes before substituting Ohm's law:

$$+6 - V_{R2} + V_{R3} = 0$$

Replace each resistor voltage with IR:

$$V_{R2} = I_2 R_2 = 5I_2, \qquad V_{R3} = I_3 R_3 = 2I_3$$

$$+6 - 5I_2 + 2I_3 = 0$$

Substitute the shared current:

$$+6 - 5I_2 + 2(I_1 - I_2) = 0$$

Distribute $+2$ to both terms in the bracket:

$$+6 - 5I_2 + 2I_1 - 2I_2 = 0$$

Reorder the current terms, then combine the I_2 terms:

$$+6 + 2I_1 - 5I_2 - 2I_2 = 0 \quad \Longrightarrow \quad +6 + 2I_1 - 7I_2 = 0$$

Subtract 6 from both sides:

$$2I_1 - 7I_2 = -6$$

Multiply the complete equation by -1:

$$\boxed{-2I_1 + 7I_2 = 6}$$

The second matrix row is $[-2,\ 7]$ and $b_2 = 6$. Multiplying a complete equation by -1 does not change its solution.

Put both equations into one matrix

5 Align the variable order

Use the same order, $[I_1, I_2]$, in every equation and every matrix row.

$$\begin{aligned} &\text{Equation 1:} \quad 5I_1 - 2I_2 = 10 \\ &\text{Equation 2:} \quad -2I_1 + 7I_2 = 6 \end{aligned}$$

	coefficient of I_1	coefficient of I_2	right side
Equation 1	5	-2	10
Equation 2	-2	7	6

$$\underbrace{\begin{bmatrix} 5 & -2 \\ -2 & 7 \end{bmatrix}}_{A\ [\mathrm{k\Omega}]} \underbrace{\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}}_{\mathbf{x}\ [\mathrm{mA}]} = \underbrace{\begin{bmatrix} 10 \\ 6 \end{bmatrix}}_{\mathbf{b}\ [\mathrm{V}]}$$

Expand the matrix product row by row:

$$\begin{bmatrix} 5I_1 + (-2)I_2 \\ (-2)I_1 + 7I_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 6 \end{bmatrix}$$

The top row is Equation 1. The bottom row is Equation 2. If either row reads back differently from the KVL equation, fix the matrix before solving.

$$[\mathrm{k\Omega}][\mathrm{mA}] = [\mathrm{V}]$$

The unit choice makes the numerical answer come out in milliamperes.

6 Use the row rule to catch transcription errors

For each row, read the matrix back as an equation before solving:

$$[5,\ -2] \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = 10 \quad \Longrightarrow \quad 5I_1 - 2I_2 = 10$$

$$[-2,\ 7] \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = 6 \quad \Longrightarrow \quad -2I_1 + 7I_2 = 6$$

Stumbling block: changing the variable order in only one row creates a different problem. The solver may still return a clean-looking answer.

Solve two ways and verify

7 Numerical linear algebra

Use a solver for $A\mathbf{x} = \mathbf{b}$. Do not calculate A^{-1} by hand for this task.

```
import numpy as np

A = np.array([[5.0, -2.0],
              [-2.0, 7.0]])
b = np.array([10.0, 6.0])
x = np.linalg.solve(A, b)
residual = A @ x - b
```

8 Symbolic solution

Symbolic algebra keeps exact fractions instead of decimal approximations.

```
import sympy as sp

I1, I2 = sp.symbols("I1 I2")
equations = [sp.Eq(5*I1 - 2*I2, 10),
             sp.Eq(-2*I1 + 7*I2, 6)]
exact = sp.solve(equations, (I1, I2), dict=True)[0]
```

The exact solution is

$$I_1 = \frac{82}{31}\,\text{mA}, \qquad I_2 = \frac{50}{31}\,\text{mA}, \qquad I_3 = I_1 - I_2 = \frac{32}{31}\,\text{mA}.$$

9 Verify independently

$$\text{Equation 1:}\quad 5\left(\frac{82}{31}\right) - 2\left(\frac{50}{31}\right) = 10$$

$$\text{Equation 2:}\quad -2\left(\frac{82}{31}\right) + 7\left(\frac{50}{31}\right) = 6$$

$$\text{KCL:}\quad \frac{82}{31} - \frac{50}{31} = \frac{32}{31}$$

A near-zero residual proves that the calculated currents satisfy the equations entered into Python. It does not prove that the arrows, signs, units, or original circuit model were correct.

Part B: build the Question 5 matrix yourself

The goal: use the same method on Question 5 without being given its coefficients, completed matrix, or currents.

10 Define the currents and signs

In the notebook, answer before doing algebra:

1. What directions did you choose for I_1, I_2, and I_3?
2. Is your shared-current equation $I_3 = I_1 - I_2$ or $I_3 = I_2 - I_1$? Why?
3. Which source and resistor terms are rises? Which are drops?
4. Why does the shared-resistor sign differ between the two loop walks?

11 Show seven algebra lines for each loop

For **each** loop, show one move per line:

1. signed KVL using resistor-voltage symbols;
2. replace every resistor voltage with IR;
3. substitute the expression for I_3;
4. distribute through the bracket;
5. reorder into $[I_1, I_2]$;
6. combine like terms;
7. move the constant to the right side.

Your two boxed equations should have the structure

$$\boxed{a_{11}I_1 + a_{12}I_2 = b_1}, \qquad \boxed{a_{21}I_1 + a_{22}I_2 = b_2}.$$

12 Extract coefficients and prove the matrix matches

Question 5	coefficient of I_1	coefficient of I_2	right side
Equation 1	_____	_____	_____
Equation 2	_____	_____	_____

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}}_{\mathbf{b}}$$

Replace the six placeholders with your numbers. Read the top row back as Equation 1 and the bottom row as Equation 2. Only then use Jupyter to solve and compare with the released answer.

> The helper questions reveal the method, not the answer. Your algebra must generate the six numbers placed into A and $\mathbf{b}$.

Assignment and submission

Part A: explain how the completed Question 4 circuit becomes two equations, one matrix, and two matching solver results.

Part B: use the helper questions to derive the Question 5 equations and matrix without copying the completed answer.

The essential skill: keep the shared-current definition and variable order consistent.

The challenge: explain every sign in both loop equations instead of copying the finished matrix.

The investigation: use Python help and the official NumPy or SymPy documentation when a function or object is unfamiliar.

The verification: use residuals, substitution, and $I_3 = I_1 - I_2$.

The reflection: identify the step that changed your understanding of the two loops.

Required notebook evidence

1. Write the definitions of I_1, I_2, and I_3 in a Markdown cell.
2. Show the three signed KVL terms for Loop 1 and derive Equation 1.
3. Show the three signed KVL terms for Loop 2 and derive Equation 2.
4. State the variable order and units.
5. Build A and $\mathbf{b}$, then read both matrix rows back as equations.
6. Solve with ‘numpy.linalg.solve‘ and report the residual vector.
7. Solve the same equations with ‘sympy.solve‘ and report exact fractions.
8. Calculate I_3 and compare all three currents with the released answer.
9. Answer the notebook reflection in complete sentences.
10. For Question 5, show both seven-line rearrangements, the coefficient table, the completed matrix, and both row read-backs.
11. Solve Question 5 only after the matrix is justified; then compare with the released answer.

Progressive hints

Hint 1 - concept: the shared resistor carries $I_1 - I_2$ because the mesh currents oppose each other there.
Hint 2 - documentation: search ‘numpy.linalg.solve A shape‘ or ‘SymPy solve simultaneous equations‘.
Hint 3 - technical: use rows $[5, -2]$ and $[-2, 7]$ only after you can read them back as the two standard-form equations.

Submit

Rename the notebook exactly:

`studentnumber-lastname-kirchhoff-jupyter.ipynb`

Example: `123456789-Smith-kirchhoff-jupyter.ipynb`. Keep the ‘.ipynb‘ ending. Restart the kernel, run all required cells, save, download, reopen, and submit the notebook through the assigned classroom system.

Later, not tomorrow: the intentional keyword error, array-shape error, unit bug, parameter sweep, capacitors, and RC response.